

Prompt Confusion Phase 04 Full Checkpoint

Combined checkpoint memo for the April 10, 2026 Phase 04 full run, covering behavior, conflict probing, arbitration readouts, exploratory PCA, and the first section-attribution plus causal follow-up pass.

MAIN READ

Phase 04 is now behaviorally sane enough for mechanistic use. The full 288-row run shows valid structured outputs on every row, a strong conflict-versus-aligned residual signal, a weaker but real conflict-side arbitration signal, and a plausible section-local policy readout in `SETTINGS` mean @ layer 24. The only major unresolved item is the first causal pass, which failed at the generation surface rather than at the representation level because the harness emitted `<think>` traces instead of parseable JSON.

Run Summary

Behavior run 16474bceae4e 288 of 288 rows produced valid structured outputs.	Conflict probe 62fc89fd861b Prompt-EOS residual probe on full aligned plus conflict split.	Arbitration probe 6924c2b36f43 Conflict-only residual readout for strategy versus setting.	Attribution pass 7e71cf742002 123 conflict rows with section-local full-sequence analysis.
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This checkpoint pulls together the local artifacts in `projects/DX_TERMINAL/prompt_confusion/phase_04/` outputs:

- `behavior_audit_16474bceae4e.json`
- `analysis_full_prompt_eos_probe/62fc89fd861b`
- `analysis_full_prompt_eos_pca/f4c70fcbac6d`
- `analysis_conflict_readout_residual/6924c2b36f43`
- `analysis_conflict_readout_pca/331de685cbc1`
- `analysis_conflict_readout_router/4aeb882aea9d`
- `conflict_arbitration_stage1`

Behavioral Sanity

The purpose of the behavior audit is not strict benchmark accuracy. The question is whether the synthetic prompts produce usable, non-collapsed behavioral differences that let later analyses distinguish conflict from alignment and SETTINGS-favored from STRATEGY-favored resolution.

Splits	Valid Output Rate	Exact Expected Rate	Readout Strategy Rate	Readout Setting Rate
overall	1.000	0.5139	0.6667	0.2396
aligned	1.000	0.7083	0.8333	0.1250
conflict	1.000	0.3194	0.5000	0.3542

Behavior interpretation

The important fact is not the 0.5139 overall exact-match rate. The important facts are that every row parsed cleanly, aligned versus conflict rows separate behaviorally, and conflict rows do not collapse onto a single source. On the conflict split, the readout-side mix is 72 strategy / 51 setting / 21 neither, which is enough diversity to support arbitration analysis.

Conflict rows by family:

Family	Rows	Strategy	Setting	Neither
activity_force_observe	36	31	5	0
activity_force_trade	36	15	21	0
trade_size_force_large	36	3	18	15
trade_size_force_small	36	23	7	6

The softest family is activity_force_trade, but it is still useful for the intended task because its conflict rows respond to pressure buckets rather than collapsing:

- balanced: 3 strategy / 9 setting
- setting_favored: 12 strategy / 0 setting
- strategy_favored: 0 strategy / 12 setting

The bucket names are slightly counterintuitive in this family, but the key result is that arbitration moves with contextual pressure instead of degenerating into random outputs.

Conflict Probe

The full aligned-versus-conflict residual probe is strong under grouped evaluation by matched_pair_id.

Layer	Balanced Accuracy	Accuracy Mean	Std Dev	Selectivity
0	0.7844	0.7844	0.0379	0.3095
20	0.7883	0.7883	0.0244	0.2814

24	0.8643	0.8643	0.0366	0.3990
28	0.8612	0.8612	0.0212	0.3682
36	0.9202	0.9202	0.0416	0.4166
44	0.8366	0.8366	0.0717	0.3537

What this supports

This is a solid Level-2 representational result. There is a strong prompt-state signature of conflict itself, especially from the mid-to-late residual stream, peaking at layer 36 with grouped balanced accuracy 0.9202.

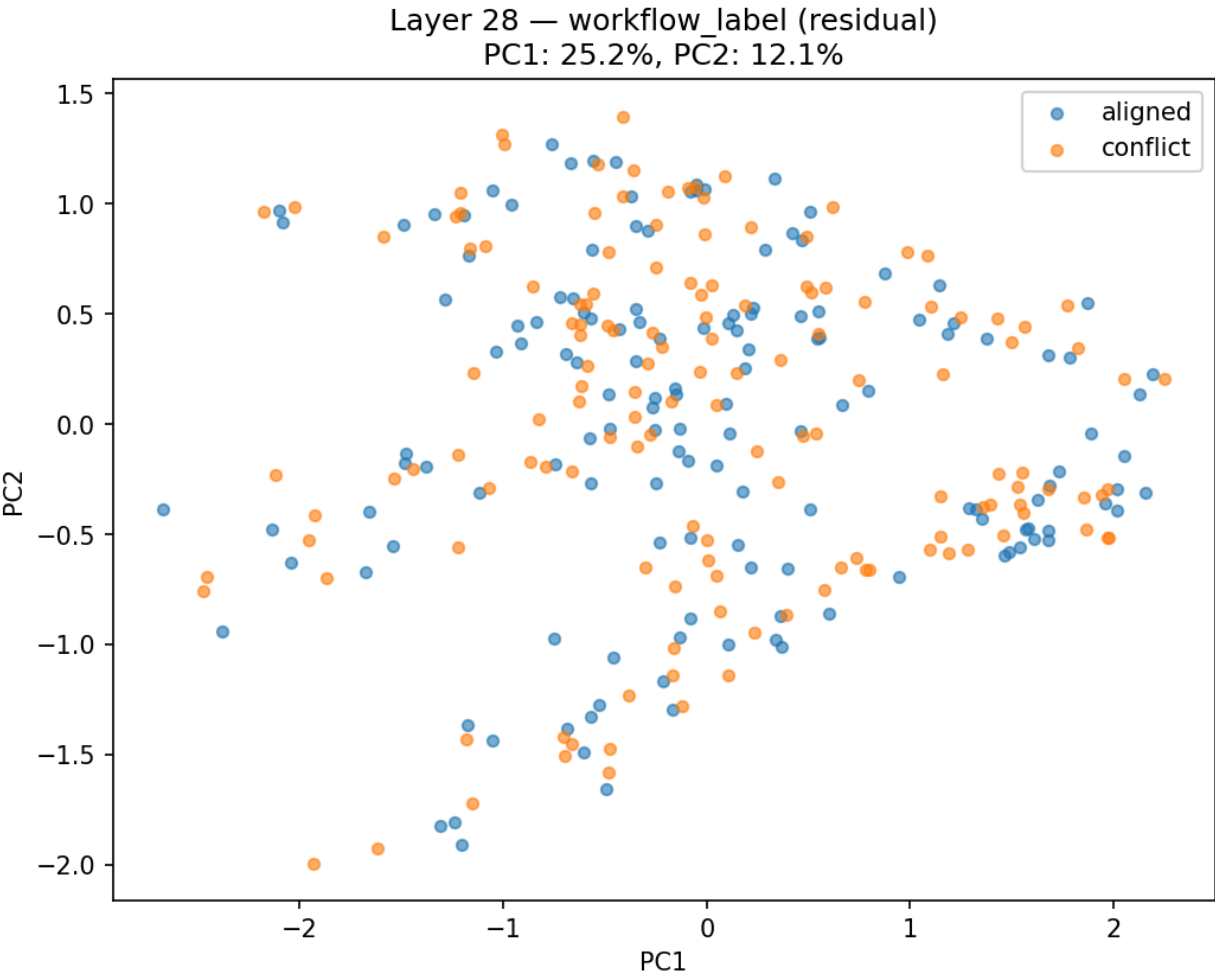


Figure 1: Exploratory PCA view for the full conflict-versus-aligned readout at layer 28. This is descriptive only, but it is consistent with the strong linear probe result in mid layers.

Conflict-Side Arbitration

The harder question is not whether conflict exists, but whether the prompt state carries information about which side of the conflict is winning. For that conflict-only readout, the residual stream remains useful and the router stream is weaker but still above baseline in a few mid-to-late layers.

Readout	Layer	Balanced Accuracy	Std Dev	Note
residual	20	0.7124	0.0621	Peak grouped arbitration readout.
residual	21	0.7075	0.0832	Nearly tied with the peak layer.
residual	41	0.6995	0.0786	Late residual still carries policy-side information.
router	16	0.6476	0.0688	Best router layer.
router	28	0.6468	0.0866	Comparable mid-layer router readout.
router	44	0.6215	0.1166	Late router signal remains above baseline.

Arbitration interpretation

Arbitration is materially weaker than the easier conflict-versus-aligned readout, which is exactly what we should expect. The residual peak of 0.7124 says there is real prompt-state information about whether SETTINGS or STRATEGY is winning. The router peak near 0.648 is weaker, but it is still enough to justify keeping router analysis in the follow-up stack.

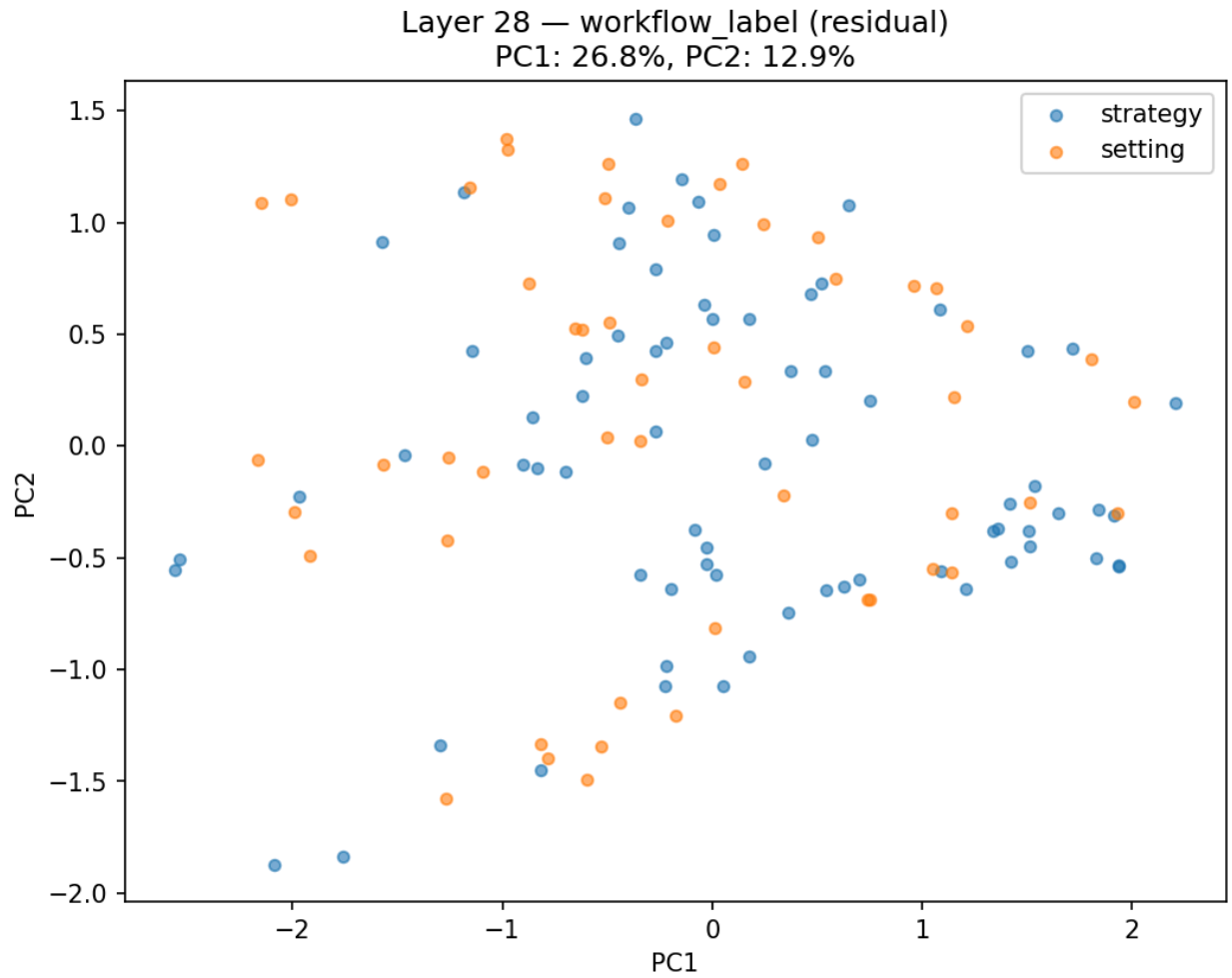


Figure 2: Exploratory PCA view for the conflict-only strategy versus setting readout at layer 28.

Section Attribution And Causal Status

The first full-sequence attribution pass asked where the cleanest policy-source readout lives once conflict rows are isolated.

Section	Layer	Balanced Accuracy	Interpretation
PORTFOLIO	28	0.7924	Best absolute residual section readout, but not the cleanest policy-source section.
MARKET	24	0.7844	Strong contextual-pressure confound candidate.
SETTINGS	24	0.7647	Best policy-source readout and selected direction.
SETTINGS	36	0.7639	Prompt-end section anchor, nearly tied.
STRATEGY	36	0.7426	Clear strategy-side readout, slightly weaker than settings.

The selected intervention target for the first causal sweep was therefore:

- section: SETTINGS
- pooling: mean
- layer: 24
- direction norm: 1.499601

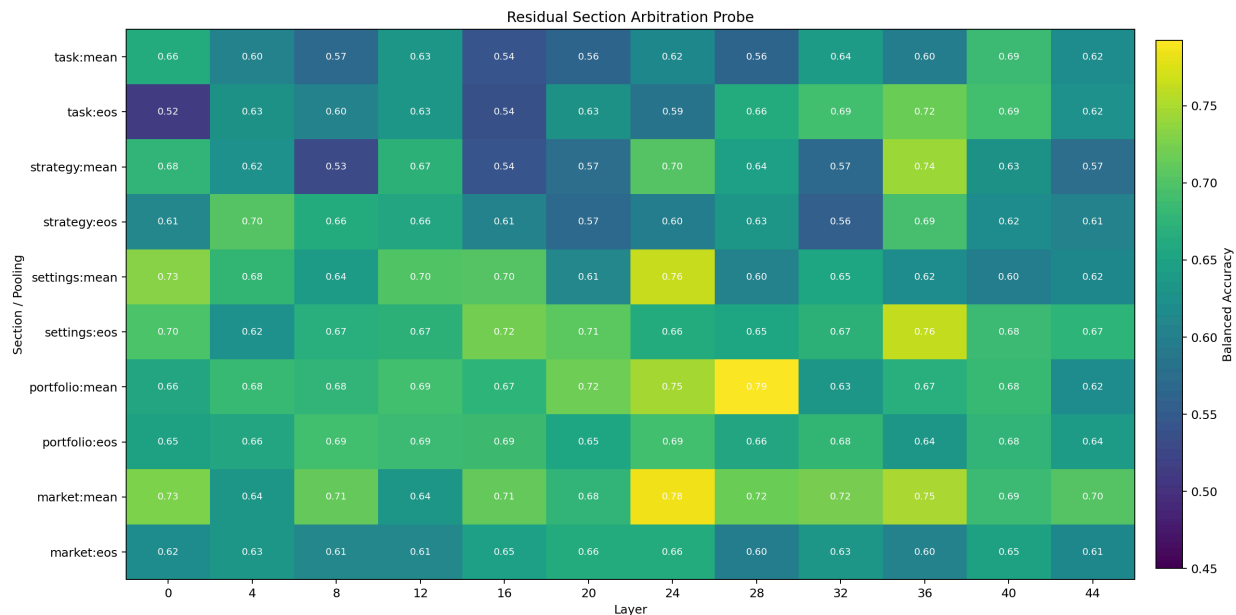


Figure 3: Grouped balanced accuracy by section, pooling target, and captured layer for the conflict-only section-attribution pass.

Two cautions matter here:

- only 25 rows had usable full-span router coverage in this full-sequence attribution setup, so the router section sweep is not interpretable yet
- the first causal generation sweep is not a valid null result, because generation returned <think> traces instead of the expected JSON action objects

Causal checkpoint

All three causal conditions came back as `neither_rate = 1.0`, but the row-level artifacts show invalid generations beginning with <think> Okay, let's break this down That means the failure was at the output surface, not obviously at the intervention site. The correct next move is to rerun the same causal setup with thinking explicitly disabled and then compare baseline, `setting_push`, and `strategy_push` by `readout_side` and by family.

What Is Supported Now

SUPPORTED

The synthetic prompts are behaviorally usable. Conflict is strongly represented in the residual stream. Conflict-side arbitration is linearly

NOT SUPPORTED YET

A successful causal intervention is not established yet. Router section localization is not established yet. The first section-attribution

decodable above baseline. A plausible policy-source readout exists in SETTINGS mean @ layer 24.

pass also warns that PORTFOLIO and MARKET carry very strong arbitration-adjacent signal, so confound control still matters.

Local Artifact Index

Primary local files for report generation and follow-up:

- projects/DX_TERMINAL/prompt_confusion/phase_04/outputs/behavior_audit_16474bceae4e.json
- projects/DX_TERMINAL/prompt_confusion/phase_04/outputs/analysis_full_prompt_eos_probe/62fc89fd861b/results.json
- projects/DX_TERMINAL/prompt_confusion/phase_04/outputs/analysis_full_prompt_eos_pca/f4c70fcbac6d/pca_layer28_workflow_label.png
- projects/DX_TERMINAL/prompt_confusion/phase_04/outputs/analysis_conflict_readout_residual/6924c2b36f43/results.json
- projects/DX_TERMINAL/prompt_confusion/phase_04/outputs/analysis_conflict_readout_pca/331de685cbc1/pca_layer28_workflow_label.png
- projects/DX_TERMINAL/prompt_confusion/phase_04/outputs/analysis_conflict_readout_router/4aeb882aea9d/results.json
- projects/DX_TERMINAL/prompt_confusion/phase_04/outputs/conflict_arbitration_stage1/section_attribution/summary.json
- projects/DX_TERMINAL/prompt_confusion/phase_04/outputs/conflict_arbitration_stage1/section_attribution/section_probe_results.parquet
- projects/DX_TERMINAL/prompt_confusion/phase_04/outputs/conflict_arbitration_stage1/causal_check/summary.json
- projects/DX_TERMINAL/prompt_confusion/phase_04/outputs/conflict_arbitration_stage1/causal_check/row_level.parquet

Next Step

The clean next step is now narrow:

1. rerun the causal harness with thinking disabled so the outputs match the JSON-only behavioral surface used elsewhere in Phase 04
2. keep the same current intervention target unless that rerun disproves it: SETTINGS mean @ layer 24
3. after the causal rerun is usable, move into attention and attribution follow-up conditioned on the conflict-side labels
4. keep explicit confound checks in frame, especially section length, family identity, and the strong PORTFOLIO / MARKET readouts

Prompt Confusion Phase 04 Full Checkpoint - generated from local outputs under projects/DX_TERMINAL/prompt_confusion/phase_04/outputs, including behavior run 16474bceae4e, conflict probe 62fc89fd861b, arbitration probes 6924c2b36f43 and 4aeb882aea9d, exploratory PCA runs f4c70fcbac6d and 331de685cbc1, and the section-attribution checkpoint downloaded to conflict_arbitration_stage1.